

Creating open educational resources



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Introduction

This course introduces you to the concepts of:

- open educational resources (OERs)
- issues involved in the creation, use and re-use, and pedagogy of OERs
- a range of tools and media to support you in developing your own teaching and learning practices.

It will provide you with the skills and confidence to engage in further OER work as both creator and user.

Find out more about studying with The Open University by [visiting our online prospectus](#)

Learning Outcomes

After studying this course, you should be able to:

- state personal motivation for producing and using OERs
- evaluate some examples of educational resources for active open learning
- plan a structured learning experience using a range of resources
- produce, release and use OER
- understand how to evaluate teaching resources.

Quiz

Welcome to the open educational resource (OER) quiz. By embarking on this course you are no doubt already considering that OER may benefit your own teaching and learning practices; the purpose of this quiz is to start you thinking about the wide range of themes, tools and resources available to those who wish to engage with OER. The quiz consists of a range of multiple choice and free text questions. The quiz should take between 30 and 45 minutes to complete.

Round 1: Images

Identify the Creative Commons licences

Question 1

What does the Creative Commons licence mean for each of these images?



Figure 1 A trumpeter

[dsbnola](#)

Provide your answer...

Answer

This picture has an **Attribution (CC BY)** licence. This lets others distribute, remix, tweak and build upon your work, even commercially, as long as they credit you for the original creation. This is the most accommodating of licences offered and is recommended for maximum dissemination and use of licensed materials.

Question 2



Figure 2 A dog
[zmtomako](#)

Provide your answer...

Answer

This picture has an **Attribution-ShareAlike (CC BY-SA)** licence. This licence lets others remix, tweak and build upon your work, even for commercial purposes, as long as they credit you and license their new creations under the identical terms. This licence is often compared to 'copyleft' free and open source software licences. All new works based on yours will carry the same licence, so any derivatives will also allow commercial use. This is the licence used by Wikipedia, and is recommended for materials that would benefit from incorporating content from Wikipedia and similarly licensed projects.

Question 3





Figure 3 A flower

[Sen](#)

Provide your answer...

Answer

This picture has an **Attribution-NonCommercial-NoDerivs (CC BY-NC-ND)** licence. It is the most restrictive of the six main Creative Commons licences, allowing others to download your works and share them with others as long as they credit you, but they can't change them in any way or use them commercially.

Name the significant locations

Question 4

What is the significance of these locations for OERs?



Figure 4 The Open University

Janet Dyson

Provide your answer...

Answer

The Open University has put more than 600 free online free courses on its website.

Question 5



Figure 5 Cape Town

[Charlie Dave](#)

Provide your answer...

Answer

Cape Town is the home of the [Cape Town Open Education Declaration](#).

Question 6



Figure 6 Stata Center

[Tjeerd](#)

Provide your answer...

Answer

This is the Stata Center at the Massachusetts Institute of Technology (MIT). In 2001, MIT was the first university to work on putting many of the teacher-defined support materials from its undergraduate and graduate courses online, in [MIT OpenCourseWare](#).

Define the symbols

Question 7

What do each of these logos symbolise?



Figure 7 A rights logo

Provide your answer...

Answer

Copyright.

Question 8



Figure 8 A rights logo

Provide your answer...

Answer

Copyleft.

Question 9



Figure 9 A rights logo

Provide your answer...

Answer

To remix.

Question 10



Figure 10 A rights logo

Provide your answer...

Answer

To share.

Round 2: Acronyms

Define the acronyms

Question 1

What do the following acronyms stand for?

OER

Provide your answer...

Answer

Open educational resource

Question 2

SCORE

Provide your answer...

Answer

Support Centre for Open Resources in Education

Question 3

JISC

Provide your answer...

Answer

Joint Information Systems Committee

Question 4

OCWC

Provide your answer...

Answer

OpenCourseWare Consortium

Question 5

VLE

Provide your answer...

Answer

Virtual learning environment

Question 6

VUSSC

Provide your answer...

Answer

Virtual University for Small States of the Commonwealth

Question 7

OERu

Provide your answer...

Answer

Open educational resources university

Question 8

Jorum

Provide your answer...

Answer

A [repository of learning and teaching materials](#), but not an acronym. The word 'jorum' is of biblical origin and means a collecting (or drinkning) bowl.

Round 3: Facts and figures

Answer these questions**Question 1**

Answer the following questions in the space provided.

What was announced in 2001?

Provide your answer...

Answer

[MIT's OpenCourseWare](#), which at the time of writing has more than 1900 courses available freely and openly online for anyone, anywhere, to adapt, translate and redistribute.

Question 2

How many members are in the OpenCourseWare Consortium? And how many open courses must they each provide?

Provide your answer...

Answer

Today there are more than 200 members, each of which has agreed to make at least ten courses available in open form.

Question 3

How many of MIT's OCW free courses have been downloaded from iTunes U?

Provide your answer...

Answer

3.7 million.

Question 4

How much has the British government spent on OERs?

Provide your answer...

Answer

£12 million.

Question 5

How many visitors have there been to the OU's Open Research Online since its launch in 2006?

Provide your answer...

Answer

More than 1.2 million.

Question 6

How much would an OER university degree cost?

Provide your answer...

Answer

According to a 2011 article on [the Times Higher Education's website](#), Wayne Mackintosh (director of the Open Education Resource Foundation) said that an OER university degree could cost '10–15 per cent' of a traditional degree.

Round 4: Proprietary or open source?

Define the applications

Question 1

Please locate the following fifteen applications on a continuum from proprietary to open source.

Acrobat

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 2

Android

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 3

BBC iPlayer

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 4

Dreamweaver

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 5

Drupal

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 6

Facebook

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 7

Firefox

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 8

Flickr

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 9

Google Docs

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 10

Linux

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 11

Moodle

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 12

Norton Antivirus

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 13

OpenOffice

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 14

Photoshop

- ☐ Proprietary
- ☐ Mixed or neither
- ☐ Open source

Question 15

VLC Media Player

- ☐ Proprietary

- ☐ Mixed or neither
- ☐ Open source

Round 5: Odd one out

Wikis

Which wiki is the odd one out?

- ☐ Wikipedia
- ☐ Wikileaks
- ☐ Wikiversity
- ☐ Wikimedia Commons
- ☐ Wikibooks

Answer

Wikileaks is the odd one out.

Broadcasters

Which broadcaster is the odd one out, and why?

- ☐ BBC
- ☐ Al Jazeera
- ☐ CNN

Answer

Al Jazeera is the odd one out because it applies a Creative Commons licence.

In early 2009, Al Jazeera launched a Creative Commons Repository, a section of its website dedicated to posting videos under the CC Attribution licence. More recently Al Jazeera launched Al Jazeera Blogs, a website featuring posts written by prominent journalists and correspondents from Al Jazeera television network, all released under a CC BY-NC-ND licence.

<https://creativecommons.org/who-uses-cc>

Governments

Which seat of government is the odd one out, and why?

- ☐ The Élysée Palace
- ☐ 10 Downing Street
- ☐ The Kremlin
- ☐ The White House

Answer

The White House is the odd one out because the Obama Administration has used Creative Commons licences in a variety of ways, from licensing presidential campaign photos, releasing information on transition site [Change.gov](https://www.change.gov) via a CC Attribution

licence, to requiring that third party content posted on [Whitehouse.gov](https://www.whitehouse.gov) be made available via CC Attribution.

<https://creativecommons.org/who-uses-cc>

Copyrights

Which of these is the odd one out, and why?

- Creative Commons Corporation
- The OER Foundation
- International Intellectual Property Alliance

Answer

The [International Intellectual Property Alliance](#) is the odd one out because it is anti-Creative Commons and open source.

Repositories

Which of these correctly lists four repositories containing OERs?

- Humbox, Flickr, Gemstone, Knowledge Cloud
- Knowledge Network, EduTube, Jorum, Sky
- Jorum, OpenLearn, MITOpenCourseWare, MERLOT
- OpenLearn, Humbox, Walk, Flickr

Answer

The correct answer is 'Jorum, OpenLearn, MITOpenCourseWare, MERLOT'.

1 What is open learning and why OERs?

Names quickly become loaded: distance learning, supported self-study, computer-based training/computer-aided instruction, home study and flexistudy, to name but a few, have all been used to describe self-instruction or self-study and many of these terms are thought wanting. The UK Open University is sometimes described as a 'distance learning institution', yet the support that students receive from their tutor through telephone, email and face-to-face tutorials, and through correspondence tuition by commenting extensively on assignments is often greater than a student receives at a 'conventional' bricks-and-mortar university. The Open University prefers to use the term 'supported open learning', and you can find out more about its approach at [the OU's study pages](#). Furthermore, the use of the word 'instruction', rather than 'study' or 'learning', implies training over education and a narrower focus.

Similarly, 'open educational resources' (OERs) as a term is often used interchangeably with – but can be distinguished from – 'open content' and OpenCourseWare.

Briefly, according to the [OpenCourseWare Consortium](#), a collaboration of more than 100 higher education institutions:

An OpenCourseWare is a free and open digital publication of high quality educational materials, organised as courses.

In 2001, MIT was the first university to work on putting many of the teacher-defined support materials from its undergraduate and graduate courses online, in [MIT OpenCourseWare](#).

The term 'open educational resources' was coined by United Nations Educational, Scientific and Cultural Organisation (UNESCO) in 2002 (Caswell et al., 2008) and it embraces OpenCourseWare but would also include any educational materials, technologies and resources offered freely and openly for anyone to use and under some licences to remix, improve and redistribute. OpenLearn is an example of a collection of OERs. The term 'open content' was first used by David Wiley, an academic now working at Utah State University and a key figure in OERs (read his open content blog [Iterating Toward Openness](#)), and the term tends to refer to all types of materials (music, video, text and so on) that are available for use under an open, 'some rights reserved' copyright licence that enables people to use, adapt and share the materials. So open content may not necessarily have an educational purpose. There are a number of different types of open licence and so the content may be 'open' but not necessarily free to use as one would like. A good review of open licences can be found on the Commonwealth of Learning website – see ['Open licenses'](#) – and this is discussed in more detail later.

Rather than spend more time looking at differences in terminology, we will now look at some examples of OERs to investigate their purpose and structure. Specifically, we will consider some different examples from this OpenLearn site. Even though a course is not a whole course, these OERs use different elements such as text, pictures and audio-visual elements that are together known as 'assets'.

Activity 1

3 hours 0 minutes

Have a detailed look at the following OpenLearn courses.

- [Play, learning and the brain](#)
- [Maths everywhere](#)

For each one, consider and write brief notes about:

- the intended learning outcomes
- the activities that learners are asked to do
- the range of media that are employed
- the teaching sequence.

Discussion

As you look through these courses you will have seen a range of activities that learners are asked to engage with. Some, such as *Play, learning and the brain*, use Flash to animate diagrams and to set up quizzes. *Maths everywhere* uses video to exemplify mathematics being used in an everyday setting and has audio clips too to talk the learner through some pictures of 'mathematical musings'.

It is clear that assumptions have been made about the intended learner. For example, *Play, learning and the brain* was written for a teacher or helper working in something

like a nursery or similar education setting, so it has a professional focus. *Maths everywhere* is from an introductory course for those adults who may feel have felt in the past that mathematics is not for them.

Activity 2

Has your institution been involved in any OER projects? What lessons can you draw on from other projects to inform colleagues and further promote your use of OER? How might you collaborate with other institutions to create and use OER?

You can complete this activity in a downloadable reflection tool, which also includes reflective questions for other topics in this course.

Section 1 resources

‘What does “open” mean in OER?’:

Video content is not available in this format.

[What is the meaning of open in OER](#)

‘Implications of OER for mediating teaching and learning opportunities – what are you trying to present?’:

Video content is not available in this format.

[The implications of OER for mediating teaching](#)

‘OERs are what people make of them’:

Video content is not available in this format.

[OER are what people make them](#)

‘What OER can do for individuals, teachers, institutions and governments’:

Video content is not available in this format.

[OER and the four major groups of people](#)

‘When might it be better to collaborate or compete in HE learning?’:

Video content is not available in this format.

[When is it better to collaborate? When is it better to compete?](#)

‘OER business models, and their sustainability and viability’:

Video content is not available in this format.

Business models and sustainability

- 'A journey through some OER projects, programmes, usage and issues' – PowerPoint presentation

2 What makes a good OER?

What is an open educational resource?

The term 'open educational resource' is one that encompasses a broad range of items. It can describe a single image or an entire short course, and materials can be in any medium or a mixture. The Organisation for Economic Co-operation and Development (OECD) has defined OERs as 'digitised materials offered freely and openly for educators, students and self-learners to use and re-use for teaching, learning and research'.

Now view the following video.

Video content is not available in this format.

[What do we mean by OER?](#)

What's so special about OER?

It is a simple idea – that you license content in a manner that explicitly encourages use and adaptation – but it has proved a very powerful one. Watch the following video for an elaboration of what's so special about OER.

Video content is not available in this format.

[What's so special about OER?](#)

Characteristics of a good OER

A good OER is:

- findable – it can be in multiple locations
- clearly described
- clearly licensed (normally through Creative Commons)
- from a source you trust
- easy to modify
- free-standing – it does not assume knowledge of other resources

- free of copyright content
- being used by/recommended by people like you
- imperfect – it just needs to work for you.

3 Finding and evaluating OERs

When seeking content for adaptation and re-use in open educational contexts there are several tools available to support discovery. Many of these tools are the result of experimental prototyping and short-term funded projects, however, and therefore carry with them a certain amount of risk. Not all are sustained beyond the life of the funding, but these initiatives have sought to use a variety of search technologies to support the discovery of generic and domain-specific OERs. As we move forward with search technologies based on increased application of [semantic approaches](#) to discovery, things should hopefully improve for the end user.

Within this course you will find a list of the current search tools available to find OERs. Many of these tools have been funded as part of the broader UK JISC OER Programme and have been supported with additional funding from the Higher Education Academies in the UK.

3.1 Finding OERs

Activity 3

Use [Google's Advanced Search](#) to find an OER in an area that interests you. It allows you to restrict your results to Creative Commons licensed material by setting the filter in the 'usage rights' field to 'free to use share or modify'. Now fill in the table supplied, which includes the list of characteristics of a good OER from Section 2, to assess how well the resource you find meets the criteria for a good OER.

Two key resources that have brought together several aspects of working with OERs and contain sections on searching and evaluating have been published by [JISC](#) and [WikiEducator](#). While these resources contain links to specific search tools that can be queried, the most effective start to finding discipline-specific OER is to query the specific open content repositories that have been built to support communities committed to working with open content.

The UK national repository for supporting work in this field is the [Jorum](#). The various jointly funded JISC/HEA projects have deposited all their outputs in Jorum and the repository continues to grow in terms of assets created and licensed for re-use for learning, teaching and research. Jorum results are now fully open.

End users searching for very specific requirements tailored to meet national or regional needs for licensed open content are frequently better served visiting and querying national repositories built to serve such needs. Exemplar repositories such as Jorum that operate within the constraints and needs of regional and national boundaries often still make their content globally open for discovery and reuse. The perceived and actual value

of this content will of course always be driven by the specific needs of the end user searching for it. Our broader global communities building, managing and repurposing this content will only ever reap the full value through users proactively engaging in feedback, enhancements and re-deposit of alternative versions, flexibly licensed for further reuse. The process of finding and evaluating OER will only ultimately improve through engagement and sharing, the key philosophy behind the movement itself.

Other exemplar repositories and digital libraries that can assist in finding specific domain OER include:

- [NSDL \(National Science Digital Library\)](#)
- [iBerry](#)
- [MERLOT \(Multimedia Educational Resource Learning and Online Teaching\)](#)
- [Academic Earth](#).

Activity 4

Visit the [JISC OER search engines page](#) and select two of the engines displayed to carry out a search on a topic of your own choosing. Compare and contrast the search experience from the viewpoint of usability as well as the quality and relevance of your results.

3.2 Evaluating open learning

Having experimented with some of the search tools available and got some results, the next step for anyone searching for relevant content is to evaluate these results in a systematic way. If you intend to use OERs for direct teaching and learning purposes, or for some repurposing prior to teaching and learning, there are several attributes that need to be considered first. Important attributes of quality OERs include:

- accuracy
- reputation of author/institution
- standard of technical production
- accessibility
- fitness for purpose
- clear rights declarations, e.g. Creative Commons.

The [JISC Open Educational Resources infoKit quality considerations](#) web page contains a range of detailed criteria for consideration. Some key criteria that you might want to adopt when evaluating your results might be grouped into attributes. For example:

- **Can the content be described as follows?**
 - Relevant, accurate, appropriate level of detail, objective, current and jargon-free.
 - Of good provenance (consider the reputation of the author/institution), with a list of references if appropriate.
 - Free of advertising.
- **Does it fit my chosen pedagogy?**

- Learning outcomes are stated and match with learner's needs.
- Engaging and interactive.
- Set at the appropriate level, with any prerequisite skills/understandings stated.
- The time required to study is stated and equates to the importance of the learning outcomes achieved.
- **How does it measure up to usability/accessibility standards?**
 - Easy-to-use and well presented, with clear navigation.
 - Accessible for users with disabilities and conforms to accessibility guidance e.g. the UK Equality and Human Rights Commission's [general web accessibility guidance](#).
- **How genuinely re-usable is it?**
 - A standalone resource that can be reused in different contexts.
 - Robust and functional, and works on different browsers/platforms.
 - Rights are fully documented, e.g. does it carry a clear Creative Commons or other rights declaration? Is it OK to re-use it? Are there any conditions?

The video below provides a record of a session previously delivered on the topic of finding and evaluating OERs as part of the activities run for SCORE fellows at The Open University. You can see from watching this that there were very diverse experiences encountered by the participants in using the search tools and locating OER that met their expectations or requirements, and that there is still a need to improve the searching experience for learners, teachers and researchers when seeking OER.

Watch the Video at [YouTube.com](#)
Finding and evaluating OERs

Lisbeth Levey highlighted the issues of finding relevant OER in a 2012 report published by the Commonwealth of Learning. Her personal account of trying to locate and use OER suitable for supporting a postgraduate university course in agriculture within an African context demonstrates:

- the richness and breadth of material available
- the considerable challenges associated with finding relevant material that is openly licensed for re-use
- the crucial need to frame searches well in order to retrieve good quality material.

Activity 5

0 hours 30 minutes

Join an OpenLearn free course that you are particularly interested in, or which you have already studied, and do the following:

- Investigate the rating system provided. Look at the questions about quality, interest and difficulty.
- Look at any constructive comments in the Comments section (if there are any).

What other questions would you wish to ask to obtain feedback on your OER?

[Discussion](#)

I would want to obtain specific feedback in addition to the general comments under the rating system. I am interested in how people have engaged with Compendium and FM, and how easy (or rather how difficult!) people find the different ways that free courses can be downloaded.

Activity 6

How do you usually evaluate sources of information for the resources that you create? Do you have to do things differently for OERs? Where can you find an OER for your discipline/subject area? What tools can you use to evaluate their usefulness?

Discussion

Creating an OER isn't vastly different from creating a normal teaching resource: it's a way of licensing a resource so that it can be shared with peers and colleagues, and enables them to change and develop it further if necessary to suit their own teaching practises. There is a wide range of repositories – some, such as Humbox for social sciences and LORO for languages, that are subject-specific. Others, such as Jorum, OpenLearn, OpenCourseWare Consortium and MERLOT, hold a wide range of resources on a variety of different subjects. There are also tools such as OpenNottingham's 'Xpert' that will help you to source content from a wide range of repositories without having to interrogate each one individually. Tools for evaluating the usefulness of an OER include:

- the credibility of the resource
- academic/professional judgement
- the licence, and whether it enables adaptation and change as needed
- the accessibility of the resource
- the platform and software the resource is built in and how easy it is to make the necessary changes (e.g. a PDF document is difficult to adapt, whereas a Word file is simple).

Only you can make the professional judgement as to whether the resource will suit your teaching and learning needs.

Section 3 resource

- 'Finding and evaluating open educational resources' – PowerPoint presentation that featured in the video above

4 Copyright and OER

I assume that you are reading this course because you would like to create a course similar to the materials that you can find on the OpenLearn website. You therefore have a teaching purpose and are particularly interested in the use of online tuition. Hopefully you

are also keen to share your teaching materials with others. But why bother creating a new OER? Surely there is so much material already available for free on the web anyway!

I would answer this in a number of ways. First: quality. You want to know that the materials that you are using yourself, or obtaining for use for others, are of high integrity; accurate and well constructed.

Second: copyright. While the copyright rules for many countries may be similar, any advice or comments given here is derived from and in the context of the UK Copyright, Designs and Patents Act 1988 ('the Act').

You may be exposing yourself or your institution to legal challenge if you use third party copyright material without permission. If you use YouTube for audio or video elements, or Flickr to store, say, a collage of pictures in a *Sgt. Pepper*-style line-up, they will remove the material from their websites if they receive a complaint.

The use of small extracts or amounts of third party copyright material in your OER is generally acceptable without attracting complaints from copyright holders. In the UK this is usually referred to as 'insubstantial'. Other countries, such as the USA, may use the term 'fair use' and provide a wider use provision within their territories. Those that are not familiar with copyright, or lack experience in working with it, may find it difficult to judge how little to use without permission. Using insubstantial parts of works needs to be decided with both a quantitative and qualitative gauge. The following may help:

- Use no more than around 400 words from a large book (**quantitative**).
- Do not take the substance of any work. For example, if you're taking quotes from a whodunit novel, do not take the part that reveals the culprit – even if only a few words (**qualitative**).
- Poetry is considered very **qualitative**, so use caution and do not take more than two lines without permission without seeking advice from someone with copyright knowledge and experience in this area.
- Music is also considered **qualitative**, so use caution, particularly with popular or well-known pieces – even if you're only using a couple of notes. Seek advice if unsure.
- Footage (film/moving images) is considered **qualitative** – so again, use caution. Footage is charged within the industry on a per second basis (frequently with 30-second and one-minute minimums being imposed for charging), so three or four seconds could be seen as a maximum insubstantial part.

Those that are more experienced in dealing with permissions and copyright may be able to apply some flexibility to the guidelines above, depending on each individual piece of content. There are other 'fair dealing' provisions in the Act (Sections 29–30) that permits use (as a defence) of copyright works, or parts, without seeking permission.

Activity 7

The following training session video explains the basics of copyright and explores some issues surrounding copyright and Creative Commons licences.

Watch the Video at [YouTube.com](https://www.youtube.com/watch?v=...)

Training video

The training video is designed to raise your awareness of copyright and other issues that may impact on your own content, or the content you may be accessing from other sites or areas for use in your OER. It may not provide all the answers you look for, but hopefully it may prompt questions and you may be alerted to raise and seek solutions before you publish your OER. There are also a few exercises in the video that are republished here, with comments that you may find useful.

If you place copyright in your OER plan and tackle all the associated considerations required for your particular content, you will not only manage the risk to your institution at an acceptable level but feel more confident about copyright. You will also feel you have done a good job and be able to move onto your next project without that feeling of unease because copyright had not been tackled as part of the overall project plan.

The video discusses:

- **Creation of copyright work:** How easy it is to create an original copyright work capable of copyright protection under CDPA 1988. (The 'birthday card'.)
- **Protection of copyright work:** No formalities (registration procedures) are required in order to get copyright works protected under CDPA 1988. Protection is given worldwide under conventions such as Berne and UCC (Universal Copyright Convention).
- **Identification and clearing of third party rights,** using content in a birthday card as an example.
- **Copyright ownership:** Exploring the issues of copyright ownership in the absence of a contract. It also looks at copyright ownership considerations involving collaborations.
- **Planning, risk and ethics:** The discussion in the video emphasises the importance of timely planning to minimise risk to your institution and consider any ethical issues that may arise, for example involving children or other vulnerable groups. Copyright does not operate in isolation. The video raises some ethics awareness for your institution such as it is not always about having the right to do – but doing the right thing!
- **Licensing Creative Commons:** The characteristics of [Creative Commons](#) licensing are explained and discussed with some exercises to further illustrate how Creative Commons licences work and to clear up any misunderstanding that may prevail about them. (For example, that Creative Commons is an organisation that has created and made available a suite of CC non-exclusive licences for the licensing of copyright works without payment to the general public. Creative Commons does not give permissions on behalf of rights owners – it provides the licences for rights owners to use.)

4.1 Creative Commons licensing

When you view the video you should look at the section on why you should choose Creative Commons, which aims to illustrate the benefits of applying a Creative Commons licence to some of your institution's works and puts the easy-to-understand terms and symbols in an international context.

Creative Commons is an organisation that has created and made available a suite of CC non-exclusive licences for the licensing of copyright works without payment to the general public. Creative Commons does not give permissions on behalf of rights owners; it provides the licences for rights owners to use.

In order to clear up any misunderstanding about Creative Commons, you should be aware of some fundamentals:

- Creative Commons (CC) licences do not replace copyright. They are a suite of (non-exclusive) licences provided by the Creative Commons organisation to facilitate and encourage rights owners (including educators) in the wider dissemination of copyright works for the mutual benefit of all communities. It reduces or eliminates onerous administrative time granting permissions and deciphering complex licences, which takes place in the absence of licences such as CC.
- Creative Commons licensing is non-exclusive, which means the licence does not apply to rights owners who have made their works available and they remain in control of their copyright works to licence in other ways (for payment, for example) should they so wish.
- You have a choice of licence or licences to suit your project.
- Creative Commons licences are not educational licences but available to all rights owners. However, Creative Commons licensing is very popular (and appropriate) within educational communities.
- The licences are designed to be easily understood (without lawyers) internationally in both their wording and symbols.
- Educators can search and locate resources under Creative Commons licences.
- Creative Commons licensing is for works protected by copyright only. However, it is not appropriate for licensing software that is more suitably licensed through other public licences available on the internet, such as from the Free Software Foundation. The wraparound documentation may be licensed through Creative Commons Licensing.
- A Creative Commons licence is non-revocable. This basically means that if you change your mind on the type of licence you have put your work out under – that is fine. However, the works already accessed and being used under that licence remain valid. So ensure you consider and are happy with your content being used in this way, even if you change your mind later on.
- Fair dealing, fair use and other exceptions to copyright are preserved.
- Moral rights are preserved by rights owners.
- Please go to the [Creative Commons website](https://creativecommons.org/) for further information.

4.2 Exercises from video

Here are some questions that were explored in the video. You may want to provide your thoughts and considerations before looking at the comments.

Activity 8: exercises from the video

Exercise 1

'I've found six images on the web for use in my course-related DVD – the resolutions are fine. However they are available under a [Creative Commons Attribution, Non-Commercial licence](#). This clearance is fine for my initial use for staff and students, but we would probably eventually hope to sell the course. Should I not bother with these images and re-select?'

Discussion

Before you decide whether to keep or ditch the images originally chosen, you should consider the following:

- You would have to expend (perhaps a lot) more time looking to replace images that already have been deemed suitable.
- While the course is being sold, it is still likely not to be considered a wholly commercial, profit-making exploitation if any income is being ploughed back into the educational activities of the institution. Therefore this can be explained to the rights owners. In the majority of cases, this is likely to result in a permission with no further payment – or a lower fee, which may be more time and cost efficient than starting from scratch.
- Ensure your records facilitate the reclearing of the images.

In order to facilitate the possibility of priorities changing during development of a course or afterwards, please ensure you keep records of the sources of your images. This will mean you can easily revisit those records to check on rights owners and find email addresses, etc. If you are unable to locate a rights owner after conducting a reasonable search, you may decide to carry on using an image on a risk level acceptable to your institution. If any rights owners get in touch, you can negotiate a suitable fee with all the information to hand (or in a place agreed from beginning). In exceptional circumstances this may result in the removal of an image – but that should only happen in exceptional circumstances, particularly if the images originally chosen are non-contentious.

If after due consideration you decide to reselect the images, then that is fine, providing you have considered all your options in relation to keeping the images or reselecting them.

Exercise 2

'We are producing a social science unit for our OER area. A colleague from the School of Health and Social Welfare has provided some lovely images of children they took while on holiday. I'm assuming that because the colleague is a University member of staff, these images will be OK to use in our OER area?'

Discussion

This is where you come across copyright issues interacting with ethics. It is likely that your institution has an ethics policy in relation to using children's images online that may provide some additional guidelines. Some areas of consideration are:

- Does copyright in the photo rest with the colleague? It will if the taking of it does not fall within their terms and conditions of employment. If that is the case then it is good practice to get a formal permission agreement for use by your institution. An email with the correct wording may suffice.

- Did the colleague obtain permission from the children's parents or guardians before the photographs were taken? Was it explained to the parents or guardians how the images may be used? If the colleague just took the photographs on holiday without permission, then this process may deem them unusable by your institution.
- Are the images of the children suitable? You should have a 'suitability gauge' for your project. You should be wary of using full face, close-up images, or images where children are perhaps posing or standing in a way that may be inappropriate to an open or wide audience. Seek advice if you are unsure (although being unsure may be an alert to choose something else).

Exercise 3

I've found an article on the web that would be brilliant for my learning object, which is intended for open use. I've tried to contact the author twice and have been in touch with the webmaster of the site to see if they can help, with no response. I've amended it, because I didn't agree with some of the points that the author was making – I think I've improved the work, actually – although obviously I left their name on it. Since I've had no response, I'm just going to use it anyway. Everyone's always talking about risk – so I'll take one. Is this OK?

Discussion

Every business or institution should operate within an acceptable level of risk. To try to operate within a zero-risk policy is not practical and will make you feel that you are not taking your business forward. However, taking risks must be enshrined within good practice guidelines. If you don't have good practice and are taking risks, then you and your institution are just being sloppy and unprofessional – and may be exposing the business to potential legal ramifications. Before you use a third party work without formal permission, you should consider the following:

- Did you contact the rights owner a few times without response?
- Did you check before your second chase-up that you had the right address? Did you try and find a telephone number for the rights owner?
- Did you alert the editor/team that if a response is not received, that the work should not be altered beyond some basic editorial changes? Altering an original work beyond this may impact on the integrity of the work and may lead the author to claim that their moral rights (Section 80 of the Act) have been infringed.
- If the editor/team did not agree with the author's comments, have they considered doing their own commentary to the paper, addressing the areas of disagreement? This needs to be done professionally and without malice or any defamatory remarks towards the author. (That may take you into another area of law – defamation!)
- Is there any other reason that would prevent you using the work without formal permission?

If after applying those considerations you still have not heard back from the author, you may be comfortable in using the work (with no alterations, other than minor editorial ones) on an 'await claim' basis. This may be an acceptable risk for your institution. If your institution does not have a risk policy to use third party works in this way, then you need to build that into your planning to ensure that you alert the team/author in good

time so that an alternative can be sourced and cleared. Please ensure you credit all rights owners whether you have had formal permission or not.

Exercise 4

'My institution has an online open learning resource and is based in the United Kingdom. We have selected an England and Wales UK licence for the use of our content. However, we have been asked by a user in China if the Creative Commons licence still applied? Does the Creative Commons licence refer to where the content is being used or where it is hosted?

Discussion

Creative Commons licences are designed to work internationally. Rights owners may choose a licence that applies to the jurisdiction (territory) where they reside. This makes it easier to apply the licence in the home jurisdiction, although this will not prevent rights owners from seeking remedy (enforcement) abroad should that become a necessity. The CC licence, irrespective of jurisdictional licence chosen, works internationally, and refers to where the content is being used.

Exercise 5

'My institution is putting some of its course materials online under a Creative Commons Non-commercial Sharealike licence. Do our logos and trademarks also form part of the Creative Commons licence terms?'

Discussion

No. Creative Commons only provides licences for copyright works that are protected under copyright law; logos and trademarks have their own separate protection under trademark law. However, if you are not sure, it is fair to assume that many users may be similarly confused. You should therefore clarify and reinforce this through your site's terms and conditions, and state quite clearly that your logos and trademarks are not part of the CC licensing. You may want to consider how your trademarks may be used within the licensed content – for example, only in unaltered content?

Exercise 6

'Do I need to choose only one type of Creative Commons licence or can I choose more than one?'

Discussion

You should consider this when developing the content for your OER or Creative Commons licensing. You are not confined to one type of licence; you can choose whichever is appropriate to your content. For example, while promoting the sharing and developing of content, there may be some content that is not appropriate for a CC sharealike. It may be sensitive (but not restrictive) and contain issues that your institution considers worth sharing. You therefore may be of the view that this content needs to be shared but in context only, and a non-commercial non-derivative licence may be appropriate. The main point is to widen your thought processes in relation to licensing to ensure that your content reaches the intended audiences in ways that retain the integrity of the content and your institution.

4.3 OpenLearn

The material here on OpenLearn has been cleared for use using the Creative Commons Attribution Non-Commercial ShareAlike 2.0 for England and Wales.

In short, this means you are free to:

- copy, distribute, display and perform the work
- make derivative works

as long as you follow these conditions:

- **attribution** – you must give the original author credit
- **non-commercial** – you may not use this work for commercial purposes
- **ShareAlike** – if you alter, transform or build upon this work, you may distribute the resulting work only under a licence identical to this one.

If you want to, look at the [Creative Commons Legal Code](#).

All third party materials in OpenLearn are made available for use in accordance with permissions granted by rights owners and are not subject to Creative Commons licence. All users are required to read terms and conditions, and any restrictions that are placed with acknowledgements. So therefore it is possible to combine Creative Commons licence terms for your own materials and other terms for third party materials to ensure your users derive educational benefit from a variety of sources.

You can find the Plain English version of this licence by clicking on the log at the bottom of this page.

My final point about why you might want to construct your learning as an OER is that in serving the needs of your own learners you will be serving the needs of other learners. It is possible to use copyright material for online use if you point users to where it can be freely accessed, but cannot be copied or altered or if you password protect it so that, in effect, you are using it just for the few learners in your class. Many institutions currently do this using Blackboard or WebCT (now owned by Blackboard and being phased out) – but what a missed opportunity! If you have put in the work to create some online learning, why not let as many learners as possible share the experience?

To summarise, the following advantages and disadvantages need to be considered before you begin to create OERs:

Advantages to OER

- Freedom of access, both for yourself and others.
- Freedom from proprietary systems and corporations.
- Encourages pedagogical innovation.
- Lowers costs to students.
- Potential publicity.
- Contribution to a community.
- Method of collaboration.
- Helpful to future educators.
- Potentially beneficial to developing nations.

- Avoids 'vendor lock-in' or a situation in which you have to use one company's products.

Disadvantages to OER

- Varying degrees of time commitment.
- Teachers sometimes not rewarded by the system for their efforts.
- Starting large projects can be difficult.
- Some projects require startup resources.
- Quality varies.
- May not meet accessibility requirements for persons with disabilities.
- Need to check accuracy before use.
- May need a high degree of customisation (called localisation in the OER community).
- Technical requirements vary and some require you to use a particular software.
- Requires varying degrees of continual financial support.
- Licensing and obtaining copyright clearance can be difficult.
- Some institutions may be concerned about 'giving it away'.

(From [WikiEducator's OER handbook for educators](#).)

Activity 9

What are the key licensing considerations you need to take into account for the resource you're working on? What is your institution's policy on Creative Commons licences? What type of Creative Commons licence would you feel most comfortable publishing my work under and why?

Section 4 resources

- 'SCORE copyright workshop' – PowerPoint presentation featured in the video earlier in this section

5 The pedagogy of open learning

One of the key differences between open learning, where the 'student' is remote from the teacher, and a learner just reading a textbook or looking up information for themselves on the internet, is the need to encourage *active* learning. Whether the material is text, online quizzes or audio-visual elements, the learner should not be a passive absorber of information but actively interacting with the resources. This is grounded in views of how people learn. But I have made some assumptions here and maybe you disagree with me.

5.1 How do people learn?

That seems a straightforward question, but you will already know from your work in producing teaching materials elsewhere that an answer is far from obvious.

Activity 10: take about two coffee breaks

If you work in a teaching context, ask a few colleagues the questions:

- How do you think people learn?
- What should we do as teachers to help that happen?

If you are not working in such a professional environment, you could ask the same questions of friends and especially parents of young children that you might know.

Discussion

Asking these questions in such a blunt way is likely to have elicited either a flippant response or maybe a cautious one along the lines of 'Everyone learns in different ways. It depends who they are. I teach depending on the needs of the student.' And so on. It is almost certain that your straightforward question did not get a straightforward answer!

All teachers, and parents for that matter, have a 'theory' of learning. It may link to formal ideas but is more often not something grand or grounded in careful research, but rather is a collection of day-by-day assumptions about what we, as teachers, should do to help those we are teaching to learn. New ideas about learning are developing and we need to test them against our knowledge of learner behaviour and the views we currently hold.

As you will probably know from your questioning of colleagues, the following are some views that people hold about how people learn:

- Knowledge and skills can be broken down into component parts and it is the teacher's job to do this for the learner. The teacher then teaches each element and gives the student sufficient repetition until the learner can give a 'positive response'. The student will generally receive the same instruction as everyone in the class, but if assessment shows that the student requires further help, then an additional programme with smaller steps over a longer time scale will be provided.
- A learner constructs meanings by getting to grips with the particular problems in hand. Private problem solving is very important and a teacher should provide the necessary stimulus material and opportunities for the individual student to learn something new. A student will not progress without plenty of practice in the activities that have already been mastered. In particular, a child will only be able to 'get' an idea when they have reached a certain stage of maturity; the teacher's job is to be aware of that and to decide when the learner is 'ready' to move on. Some learners are never able to 'get' certain ideas.
- All learners are educable and are helped in their learning by discussion and other social interaction, including with a more experienced learner or teacher. There is no fundamental difference between the learning of children and that of adults. Rather than waiting for a student to be 'ready' to learn, a teacher is finding out what the learner thinks in order to guide and support what the learner is trying to do next. By

talking with the teacher, and obtaining other support, a learner is able to grasp ideas and new understandings that they could never arrive at on their own.

These very brief summaries relate to the three main traditions of learning theory: behaviorism, Piagetianism and social constructivism.

5.2 Behaviorism, Piagetianism and social constructivism

How do the well-known ideas of behaviorism, Piagetianism and social constructivism relate to what you actually do as a teacher in a face-to-face context? Are you able to 'sign up' to any one of the theories wholeheartedly? As you read the descriptions you may have felt that each of them separately described some aspects of your ideas about learning and those of your colleagues, yet none was wholly satisfactory in its own right. For example, in teaching certain practical skills, a regime of practice and reinforcement in the 'behaviorist' tradition may be appropriate. An individual project will provide problem-solving opportunities and will be successful if the learner is working largely within his or her capabilities; a Piagetian standpoint. That teaching methods should be selected in terms of 'fitness for purpose', rather than adherence to a particular dogma of 'good practice', is clear. Teachers tend to have their preferred way of working, which reflects a personal 'theory', but nevertheless are not hidebound by particular ideologies and will adopt a different teaching strategy if they think it will be helpful. Sometimes it is called a 'folk theory' of learning.

Some people think that good teaching means the same thing as good explaining – keep it clear and simple and all will understand. In fact some teachers get very upset when, despite their greatest efforts, the learners just don't grasp what they have explained. When students just don't 'get it' they take it as a personal failure, or maybe blame the learners themselves.

It is certainly true that a key teaching skill is the ability to explain and describe things clearly. But a belief that transmitting information clearly is all that is required for a 'good' teacher is insufficient. However, such a 'folk' theory of how minds work is very common, and also explains the position some parents take to learning and teaching. These common beliefs were investigated by Bereiter and Scardamalia (1996), who characterised a folk theory of mind as follows:

- knowledge is 'stuff'
- the mind is a container
- learning involves putting stuff in the container

This tends to be reinforced by national curricula and examination syllabuses, which emphasise content knowledge. Bereiter and Scardamalia suggest that the corollaries of such a view of the mind is:

- pedagogy: a craft for stocking minds
- educational testing: a process for inventorying mental contents.

Desforges (2001) indicates that the corresponding 'folk pedagogy' to such a view of learning has had some remarkable success in teaching through 'show and tell'.

But where the 'stuff' metaphor breaks down – as it does with wisdom, creativity, knowledge creation, appreciation, a 'feel' for a subject, we are left floundering.

(Desforges, 2001, p. 25)

Folk theories are indeed robust, yet the alternative ideas about teaching and learning outlined above have been considered for at least the last 50 years and, in more recent times, linked to a growing understanding about the biology of the brain.

Taking a social constructivist view of learning, the experiences that we should construct is not a 'lecture' but rather a one-to-one 'tutorial'. How would you behave in those different contexts?

Activity 11

0 20

Imagine you have just one learner and you were going to work with them for about three hours to help them learn a key idea in your subject area.

You are planning what to do and it is just you and the learner.

In broad terms – What would you plan to do? How would you characterise the activities and the way you would work?

Discussion

To begin with I expect that you would not be planning to talk at length as, maybe, you would do if it was a big group that were expecting a 'lecture'. You would make sure that the learner was with you, stopping and asking for feedback or for their own ideas. You might ask for examples from the learner from their experience or you might ask them to use your idea in a new situation or practice what they have learnt in a new situation. You might ask the learner to evaluate an idea or compare one idea with another. In short, you would ask them to be **active**.

Even when you are writing educational text, write it as though you are writing an interactive tutorial. Build in activities for the learner to do that will help them to learn. Activities such as questions, tasks and exercises are a very important feature of self-instructional material as they challenge the learner to do something they can assess and appreciate for themselves. If we take the view that we construct knowledge, then using the ideas that we are learning is vital. Remember the proverb often ascribed to the Chinese:

I hear, and I forget;

I see, and I remember;

I do, and I understand.

In addition to activities, however, is the style of writing. The one-to-one tutorial will be intimate and conversational and that is the style of writing that engages the learner. You should be able to 'hear' the writer talking to you. Of course, ideally it will not only be the writer contributing to the learner's engagement with the materials. As I described earlier, learning can be enhanced if we recognise that it is a social activity, too, involving interactions with other people, family, friends and work colleagues as well as fellow learners.

6 Tools for creating an OER

In planning your learning resource, you need to keep four questions in mind.

- 1 What are you trying to achieve with this teaching course – what are your aims?
- 2 What activities do you wish the learners to engage with in order to demonstrate or achieve those aims – what are the learning objectives or outcomes, and how are they to be assessed?
- 3 How will you evaluate the effectiveness of what you have produced?
- 4 In the light of the evaluation, how will you change 1 and 2 above?

When you meet your learners face-to-face, it is relatively straight forward to answer question 3. Even before an assignment is marked or any informal question is answered, the learner's body language gives plenty of feedback as to the success of your activities. In producing OERs you don't have such quick feedback, although some research into the The Open University's OERs has taken place.

Watch the video below for a brief introduction to creating an OER.

Video content is not available in this format.

[Creating an OER](#)

6.1 What are aims and objectives/outcomes?

It is best to start to settle on the aims and objectives/outcomes (these terms are variously used around the world but are largely interchangeable) of your free course as soon as possible. You looked at the intended learning outcomes of some courses in Section 1. The difference between aims and objectives is that the aim is the general statement of what you hope the course will achieve, usually expressed in terms of what you will be presenting in the course; the objectives are what you intend the learner to be able to know, understand and do once they have studied the course. For example:

Aim: To explain the concept of energy and the need to conserve heat in houses.

Objectives: On completing the course the learner should be able to:

- list different forms of energy
- describe how energy may be transformed from one form to another
- describe different forms of heat flow
- distinguish between energy and power
- etc.

Writing learning objectives can be quite demanding, because they have to be set at the right levels of difficulty and detail for the expected learners and be reasonably assessable. Derek Rowntree (1986, p. 45) suggests the words in Table 1 when writing objectives.

Table 1 Words to avoid and use

Avoid words such as:	Use words such as:
know	state
understand	describe
really know	explain
really understand	list
be familiar with	evaluate
become acquainted with	identify
have a good grasp of	distinguish between
appreciate	analyse
be interested in	outline
acquire a feeling for	summarise
be aware of	represent graphically
believe	compare
have information about	apply
realise the significance of	assess
learn the basics of	give examples of
obtain a working knowledge of	suggest reasons why

The list on the left shows unobservable states of mind (i.e. very difficult to assess), whereas the list on the right is more focused on what the learner is able to demonstrate to others.

It is quite unlikely that in creating your OER, you will start with a blank sheet of paper as some of the original authors did on OpenLearn.

For you, the model is likely to be that described by [David Wiley](#). He believes that OER development follows a life cycle like this:

- 1 **Get:** Searching and finding OERs. Getting OERs may include using search engines, repositories and finding individual websites.
- 2 **Create:** Generate the OER, preferably using open source tools.
- 3 **Localise:** Essentially, localising means making a resource more useful to a particular situation. For example, translating instruction from one language to another.
- 4 **Remix:** Remixing is the act of taking two OER materials and merging them to form a new OER. Remixing is probably one of the most enjoyable parts of OER production.
- 5 **Licensing:** License the work using an open content licences such as Creative Commons and GFDL (GNU Free Documentation License).
- 6 **Use:** This covers the actual use of OER for your context.
- 7 **Redistribute:** Publishing an OER once it is finished and making it available for the open education community to begin the life cycle again.

However, before finding and remixing OERs to create just what you want for your learners, you need to be certain you know what it is you want the learners to have learnt. After setting out aims and objectives, one way to help you do this is to draw a diagram.

Activity 12

2 hours 0 minutes

Look at the 'spider diagram' – sometimes called a spray diagram – that I drew on paper to bring to mind the different science activities that I had engaged in as a student of science from my primary school to higher education. It was for a masters course on *Contemporary issues in science education* (SEH806), part of which is now available as the OpenLearn course [Changes in science education](#). As well as exemplifying what I wanted the students to do, it helped me collect my thoughts about different phases of science education that I wished to consider.

You might want to draw a similar diagram to set out your ideas for your OER. (If you are unused to using diagrams, try studying either of these OpenLearn courses:

[Working with diagrams](#) or [Systems diagramming](#).)

For this you could create a knowledge map using Compendium. You can also access [a short tutorial on how to use it](#).

Discussion

The Compendium tool could be used to create an online learning scheme that is more in the control of the learner – a 'north-east quadrant' OER perhaps – where the conduct of tasks is up to the learner who can follow the route through the materials that interest them. Another use is to create south-west quadrant 'teacher determined open-ended strategic learning activities'. Here the learner has control and the learners can work in groups. An example of that would be the 'Compendium for elearning'.

6.2 Creating your OER

Sources of material

You will probably be making an OER in an area in which you have some expertise so you are likely to already have lesson plans and resources that you use in your face-to-face work that will be invaluable to others.

As well as your own materials, you might like to look at a range of other OER repositories in addition to OpenLearn.

Video content is not available in this format.

[Creating OER](#)

Activity 13: what is available already for me to use?

1 hour 0 minutes

Look at the following OER repositories where there are often not whole units but rather useful 'bits and pieces' that could be mixed (but also be careful to look at the licence used in each case). This activity should take about an hour to scan what is available. This is by no means an exhaustive list, but serves as a useful starting point for creating your own reference list:

Music

[ccMixer](#)

Multimedia resources across a range of topics

[MERLOT](#)

[Xpert](#)

Images

[Flickr](#) – this is not a specific OER repository but some of the content is available for you to use freely under a Creative Commons licence. Be sure to use the advanced search to select Creative Commons licensed content.

Video

[YouTube](#) – this is not a specific OER repository but some of the content is available for you to use freely under a Creative Commons licence. Be sure to use the advanced search to select Creative Commons licensed content.

General repositories

[UNESCO OERs](#)

[Jorum](#) – a sharing site for Higher Education in the UK

[OER Commons](#) – this site has a range of open resources

Science

[Science repositories](#)

Humanities

[Humanities repositories](#)

[HumBox](#)

Languages

[LORO](#)

Let us now explore the different types of content resources.

Content

If we look specifically at OpenLearn free courses, the content comprises both the course (structured self-study resources) as well as the individual assets that make up a course. The assets of a course are the materials such as text, images, animations, audio clips, etc., which are likely to be in different digital formats. In some cases a course will consist of just one asset, but most contain a variety.

As the number of OpenLearn free courses grows, so does the variety available. However, the main types of courses that you will encounter are:

- **Segments of current and discontinued OU courses or support materials:**
These comprise an XML-based front page providing a short description of the course, its learning outcomes, hyperlinks to the sections of the course and the

course's assets (text, audio, images, etc.). In these courses the majority of the original third-party material has been cleared and retained.

- **Bespoke materials:** These are courses that have been specially written that may not be associated with an OU course. These comprise an XML-based front page providing a short description of the course, its learning outcomes, hyperlinks to the sections of the course and the courses assets (text, audio, images, etc.). In these courses third-party material has been cleared for use on OpenLearn.

Formats

Later in 2013, OpenLearn free courses will be available to be downloaded or taken away in several formats:

- print format
- course content XML
- course content RSS
- OU XML package
- IMS Content Package
- IMS Common Cartridge
- plain zip
- Moodle back-up.

At the asset level, the major formats you will find are:

- text in XML or PDF
- animations in Flash
- images such as GIF or JPEG
- audio files such as MP3 and video files such as MP4.

To help you to rework OpenLearn free course material, the OUXML structured authoring schema is provided as part of the downloaded course – although you do need to know about authoring in XML.

Tools

There are many tools for creating OER that at a basic level can be split into those with open licensing and ease of remixing built in those that don't. You may already be using the latter to create content.

By far the most commonly used tools for creating educational resources and therefore OERs is Microsoft Word and PowerPoint. To make these resources open you will need to apply an appropriate licence where – as with tools such as LabSpace from The Open University and Xpert from the University of Nottingham – the licensing is built in and the tools you use to create and therefore remix the content is free and open so you can make this easily available to others.

There are more specialist proprietary tools such as Photoshop and Flash from Adobe, where the content you produce at the end of the process can be made open but the user would need to buy the software to easily remix it.

There is an ever-increasing list of tools available for you to create content in multiple formats; a simple Google search will return many results. When choosing a tool, you will

need to think about how you want the final output to be used. If you want your content to be changed, it will require specialist software to produce the results you require.

Video content is not available in this format.

[Choosing tools](#)

7 Making the case for OER

To some, the case for open educational resources is taken as self-evident. The internet is a great platform for sharing information at no apparent cost, so why not use it as a great platform for learning? This surface argument should not be dismissed – and indeed a ‘just get on and do it’ attitude has led to many people joining in. However, if you need to make a case for OERs, then it is useful to be able to identify the benefits for each of those involved in using OERs – the learner, the organisation and the educator. OERs offer very apparent opportunities to those who receive content without charge; on the other hand, producing or teaching with OERs involves time and effort, and so the benefit to the provider or educator can be less obvious.

In this short section you will look at some benefits that do come to those who take part in providing and using OER. The particular case that is used to provide examples is the OpenLearn initiative at The Open University, although there are many other OER actions that could also be used to show benefits. OpenLearn has an advantage that when it launched in 2006, it was designed to be an experiment to see what impact offering open resources would have on the users of the site and on the University itself. The project was careful to gather information that would help see that impact and so can provide pointers to illustrations over a range of issues.

7.1 How does learning take place in the open?

Activity 14: become a learner

Time: from 20 minutes to a lifetime

Using OpenLearn or any other source of OERs, find a topic that interests you. Now become a learner! Sign up and register or commit to starting working through the resource. This differs from asking you to review material and you may well find that you start to think about the presentation and design of material in different ways. Reflect on what you are looking for and what is keeping you going.

Does access to content bring benefits to users?

The primary aim of OpenLearn was to provide resources that were available as far as possible without any barrier to use and reuse. This meant providing free content that was accessible without any registration and with an explicit licence that allowed reuse. In

OpenLearn we know that there were many people who made use of the site, with access of the resources reaching up to 11 million unique visitors per year. A majority of these visitors only access the site for a short time, but others engage more closely with the materials.

Surveys and study of the way the site is accessed revealed some interesting approaches and motivations in the way that people use the site. Using OpenLearn is seen both as a leisure activity that is fun in itself and also as a step towards more formal courses. The appeal of different aspects of the site clustered in the data around those who identified more social connections as the attraction ('social learners') and those who were more interested in the structured learning material itself ('volunteer students'). Social learners use the content as a way to connect with other people who shared the same interests and appear less interested in working through content. Volunteer students include those who want to learn but use the open resources for reasons of time, cost and opportunity.

Activity 15: what sort of learner?

Which are you – a social learner or a volunteer student? If you were looking for content to work through, do you want structure, pace and visible goals, or freer options with no need to assess yourself and links to other people interested in the same area? Of course, these are not exclusive motivations, but they do lead in different directions. Courses that are free but based strongly on formal structures with start dates and assessment show that there are significant numbers of users that are interested in being 'volunteer students'.

7.2 Organisational benefits

For The Open University as an organisation, OpenLearn has brought a range of further benefits. There has been a steady flow of students recruited whose first exposure to OU content has been through OpenLearn. It is estimated that several thousand students have now registered with the OU through OpenLearn, generating real value in fees. In addition, inquirers who first use OpenLearn turn out to be more likely to become a student than if they came through other routes such as a press campaign, or following on from a co-produced television programme.

Other benefits can be harder to measure in financial terms

Identified within OpenLearn study, other benefits of OER have helped experimenting with technology, improved routes for collaboration, allowed testing of new elements of the curriculum, and led to further funded research or development projects.

OpenLearn benefits

The benefits of OpenLearn to the OU include:

- developing and extending the reputation of the University

- deepening and broadening the community
- contributing to the University's information, advice, guidance, outreach and widening participation activities
- lessons and benefits gained from exposing and describing the OU's content through OpenLearn
- the benefits of testing and experimenting with new technologies
- creating and nurturing strategic partnerships
- exploring, examining and improving organisational structures and processes
- enhancing and building upon research strengths.

(A more complete list of OpenLearn's benefits to the OU can be found in the [OpenLearn Research Report](#).)

The benefits described above are partly attributable to the OU being a distance learning organisation, so allowing it to attract new students directly. However, survey work carried out by MIT on its OpenCourseWare project has also demonstrated the potential for similar benefits, with a growing proportion of its users being motivated learners who use open content as a way to explore their options for further study.

Activity 16: what benefits for organisations?

The OpenLearn study shows the advantages of looking for all aspects of information from an OER programme. Planning an evaluation and looking for expected gains has so far been revealing. If you are considering an OER intervention, then spend some time to think of what aspects you might review in order to learn from the experience of offering OER. Itemise the factors and then compare them to the list in the discussion below.

Discussion

Benefits that have the potential to be measured include:

- student recruitment and retention
- new partnerships/collaboration
- activities or projects supported through OER.

More often benefits are less tangible but clearly exist:

- opportunities to experiment
- development of staff
- low-level collaborations
- enhanced reputation.

Involvement in open education can also give a good feeling of being part of an international community.

7.3 Benefits for you

In many ways the biggest challenge is whether getting involved in using open education is something that you want to do. As with any activity, it takes time and energy. If you are

involved in teaching, does the range of resources give you a better starting position than other approaches? Do the resources and approaches to pedagogy that are supported by OER suit your students and learning? If you are trying to persuade other people, are there benefits that will help the organisation to change?

Activity 17: retaking the quiz

Now that you've come to the end of this free course, you may want to retake the quiz to see how far you've come. We hope that you found the course useful and informative, and that as you go forward you will consider embedding open educational resources and practises into your own teaching and learning practice.

Final thoughts

OERs have the potential to spark radical change; they also offer ways to bring in new learners and find new ways of working. Review the reasons that fit with what you need to do and plan what you need to do to do more – if necessary, revisit the sections about identifying, finding and learning with OERs. Good luck.

Conclusion

This free course provided an introduction to studying education. It took you through a series of exercises designed to develop your approach to study and learning at a distance and helped to improve your confidence as an independent learner.

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References

Academic Earth, www.academicearth.org (accessed 26 June 2012).

Attwood, R. (2011) ‘“OER university” to cut cost of degree’, *Times Higher Education*, 10 February. Available from: www.timeshighereducation.co.uk/story.asp?storycode=415127 (accessed 23 July 2012).

Bereiter, C. and Scardamalia, M. (1996) ‘Re-thinking learning’ in D.R. Olson and N. Torrance (eds) *The Handbook of Education and Human Development: New Models of Learning, Teaching and Schooling*, New York, Blackwells.

ccMixter, <http://ccmixter.org> (accessed 24 July 2012).

Centre for Educational Research and Innovation (2007) *Giving Knowledge for Free: The Emergence of Open Educational Resources*, OECD. Available from: www.oecd.org/dataoecd/35/7/38654317.pdf (accessed 16 July 2012).

Creative Commons, <http://creativecommons.org> (accessed 26 June 2012).

Desforges, C. (2001) ‘Familiar challenges and new approaches: necessary advances in theory and methods in research on teaching and learning’, *Nuttall/Carfax Memorial Lecture, Cardiff 2000*, Southwell, British Educational Research Association.

Equality and Human Rights Commission (?) ‘General web accessibility guidance’. Available from:

www.equalityhumanrights.com/footer/accessibility-statement/general-web-accessibility-guidance (accessed 26 June 2012).

Flickr, www.flickr.com (accessed 24 July 2012).

Google Advanced Search, www.google.ca/advanced_search (accessed 26 June 2012).

Hofman, J. and West, P. (2008) ‘A draft of Chapter 6 – “Open licences” of copyright for authors, educators and librarians’, Commonwealth of Learning website, 12 April. Available from: www.col.org/resources/knowServices/copyright/Pages/openLicense.aspx (accessed 26 June 2012).

HumBox, <http://humbox.ac.uk> (accessed 24 July 2012).

iBerry, <http://iberry.com> (accessed 26 June 2012).

International Intellectual Property Alliance, www.iipa.com (accessed 24 July 2012).

Iterating Toward Openness, <http://opencontent.org/blog> (accessed 26 June 2012).

Jorum, www.jorum.ac.uk (accessed 24 July 2012).

Languages Open Resources Online (LORO), <http://loro.open.ac.uk> (accessed 24 July 2012).

Levsey, L. (2012) 'Finding relevant OER in higher education: a personal account', Chapter 8 in Glennie, J., Harley, K, Butcher, N. and van Wyk, T. (eds) *Open Educational Resources and Change in Higher Education: Reflections from Practice*, Commonwealth of Learning. Available from www.col.org/PublicationDocuments/pub_PS_OER_web.pdf (accessed 25 June 2012).

MIT OpenCourseWare, <http://ocw.mit.edu/index.htm> (accessed 24 July 2012).

Moodle, <http://moodle.org> (accessed 24 July 2012).

Multimedia Educational Resource for Learning and Online Teaching (MERLOT), www.merlot.org/merlot/index.htm (accessed 26 June 2012).

The National Science Digital Library (NSDL), <http://nsdl.org> (accessed 26 June 2012).

Obama–Biden Transition Project, <http://change.gov> (accessed 24 July 2012).

OER Commons, www.oercommons.org (accessed 24 July 2012).

Open Educational Resources infoKit, <https://openeducationalresources.pbworks.com/w/page/24836480/Home> (accessed 26 June 2012).

OpenCourseWare Consortium, www.ocwconsortium.org/index.php (accessed 26 June 2012).

Rowntree, R. (1986) *Teaching Through Self-instruction*, London, Kogan Page.

UNESCO International Institute for Educational Planning (IIEP) (ongoing) 'OER useful resources/repositories'. Available from: http://oerwiki.iiep.unesco.org/index.php/OER_useful_resources/Repositories (accessed 24 July 2012).

The White House, www.whitehouse.gov (accessed 24 July 2012).

WikiEducator, http://wikieducator.org/Main_Page (accessed 24 July 2012).

Wikipedia (ongoing) 'Semantic search'. Available from: http://en.wikipedia.org/wiki/Semantic_search (accessed 26 June 2012).

Xpert, www.nottingham.ac.uk/xpert (accessed 24 July 2012).

YouTube, www.youtube.com (accessed 24 July 2012).

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